



Vidyavardhini's College of Engineering & Technology
Department of Computer Engineering
2024-2025

Experiment No.4
To Perform analysis of volatile memory to detect evidence of attack
Date of Performance:
Date of Submission:



Vidyavardhini's College of Engineering & Technology

Department of Computer Engineering

2024-2025

Aim: To perform analysis of Volatile memory to detect evidence of attack

Objective: To make use of volatility tool and its plugin to extract the evidence from the dumped volatile memory image of Windows based computer system

Theory:

Volatility is an open-source memory forensics framework for incident response and malware analysis. This is a very powerful tool and we can complete lots of interactions with memory dump files, such as:

List all processes that were running.

List active and closed network connections. View internet history (IE).

Identify files on the system and retrieve them from the memory dump. Read the contents of notepad documents.

Retrieve commands entered into the Windows Command Prompt (CMD). Scan for the presence of malware using YARA rules.

Retrieve screenshots and clipboard contents. Retrieve hashed passwords. Retrieve SSL keys and certificates.

The volatility plugins

The volatility tool supports various plugins in order to extract the information from dumped volatile memory images of Windows, Linux and Mac based computer system. Following is a list of few plugins for working with Windows based computer system dumped volatile memory.

Registry Analysis Plugins

1. hivelist : find and list available registry hives
2. hivedump: print all keys and subkeys in a hive
3. printkey: output a registry key , subkeys and values
4. dumpregistry: extract all available registry hives
5. userassist: find and parse userassist key values
6. hashdump: dump user NTLM and Lanman hashes
7. autoruns: shows autoruns key related information



Vidyavardhini's College of Engineering & Technology
Department of Computer Engineering
2024-2025

Process related Plugins

1. pslist: High level view of running process
2. psscan: scan memory for EPROCESS blocks
3. pstree: Display parent-process relationship

Code injection related Plugins

1. malfind: Find injected code and dump sections
2. ldrmodules: Detect unlinked DLLs
3. hollowfind: Detect process hollowing techniques

Analyze process DLLs and Handles related Plugins

1. dlllist: List of loaded dlls by process
2. getsids: print process security identifiers
3. handles: list of open handles for each process

Process, Drivers, and Objects related Plugins

1. dlldump: Extract dlls from specific processes
2. moddump: Extract kernel drivers
3. procdump: Dump process to executable sample
4. memdump: Extract every memory sections into one file
5. filesan: scan memory for FILE_OBJECT handles
6. dumpfiles: Extract FILE_OBJECTS from memory
7. svcscan: scan for Windows service record structures
8. cmdscan: scan for command_history buffers
9. consoles: scan for CONSOLE_INFORMATION output

Rootkit related Plugins

1. psxview: find hidden process using cross-view
2. modscan: scan memory for loaded, unloaded and unlinked drivers
3. apihooks: find API/DLL function hooks
4. ssdt: Hooks in system service descriptor table



Vidyavardhini's College of Engineering & Technology
Department of Computer Engineering
2024-2025

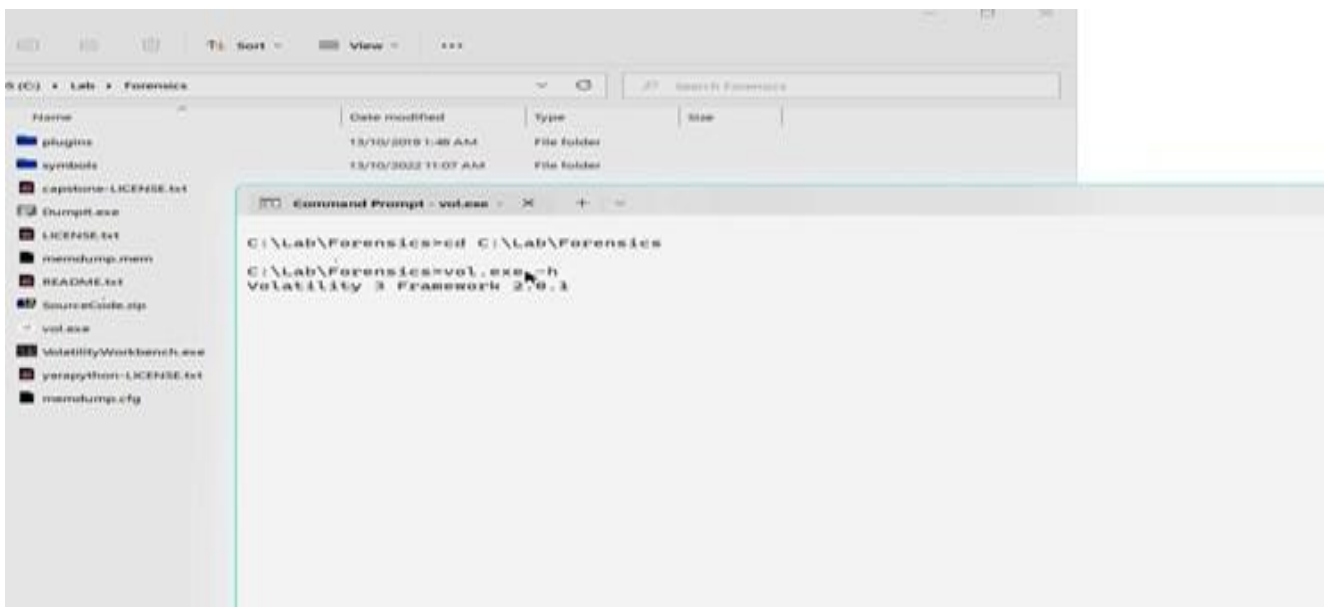
5. driverirp: identify I/O request packet (IRP) hooks
6. idt: display interrupt descriptor table

Process:

- Step 1. Create a dump of the running computer system using dump tools such as dumpIt
- Step 2. Install volatility on your computer system or copy the downloaded volatility-master directory onto a location of your computer
- Step 3. Open the command terminal
- Step 4. Go to the specific location where you copied volatility-master
- Step 5. Enter the volatility-master through command prompt
- Step 6. Use the command in the following format to work with the dumped memory image

`>> vol.py command -f dumped_file_name`

Output:





Vidyavardhini's College of Engineering & Technology

Department of Computer Engineering

2024-2025

```
Name      Date modified      Type      Size
plugins   13/10/2019 1:48 AM  File folder
symbols   13/10/2022 11:07 AM  File folder
capstone-LICENSE.txt
DumpIt.exe
LICENSE.txt
memdump.mem
README.txt
SourceCode.zip
vol.exe
VolatilityWorkbench.exe
yara-python-LICENSE.txt
memdump.cfg
```

```
Command Prompt - vol.exe - X + v
windows.registry.hivelist.HiveList
    Lists the registry hives present in a particular memory image.
windows.registry.hivescan.HiveScan
    Scans for registry hives present in a particular windows memory image.
windows.registry.printkey.PrintKey
    Lists the registry keys under a hive or specific key value.
windows.registry.userassist.UserAssist
    Print userassist registry keys and information.
windows.skeleton_key_check.Skeleton_Key_Check
    Looks for signs of Skeleton Key malware
windows.ssdt.SSDT
    Lists the system call table.
windows.statistics.Statistics
    Reads output from the strings command and indicates which process(es) each string belongs to.
windows.symlinkscan.SymlinkScan
    Scans for links present in a particular windows memory image.
windows.vadinfo.VadInfo
    Lists process memory ranges.
windows.verinfo.VerInfo
    Lists version information from PE files.
windows.virtmap.VirtMap
    Lists virtual mapped sections.

The following plugins could not be loaded (use -vv to see why): volatility3.plugins.windows.cachedump,
volatility3.plugins.windows.callbacks, volatility3.plugins.windows.hashdump, volatility3.plugins.windows.lsadump,
volatility3.plugins.windows.svcscan, volatility3.plugins.windows.vadinfo, volatility3.plugins.yarascan

C:\Lab\Forensics>vol.exe -f memdump.mem windows.netstat
Volatility 3 Framework 2.0.1
Progress: 14.24      Scanning memory_layer using BytesScanner
```

```
OS (C:) > Lab > Forensics
```

```
Command Prompt
00 0xb88fc58a45b0 UDPv4 127.0.0.1 1900 * 0 5908 svchost.exe 2022-10-13 09:23:29.0000
00
0xb88fc3772df0 UDPv4 0.0.0.0 5353 * 0 2128 svchost.exe 2022-10-13 09:23:09.000000
0xb88fc3777a80 UDPv4 0.0.0.0 5353 * 0 2128 svchost.exe 2022-10-13 09:23:09.000000
0xb88fcb136a10 UDPv4 0.0.0.0 5355 * 0 2128 svchost.exe 2022-10-13 10:02:20.000000
0xb88fcb136a10 UDPv6 :: 5355 * 0 2128 svchost.exe 2022-10-13 10:02:20.000000
0xb88fcb13b300 UDPv4 0.0.0.0 5355 * 0 2128 svchost.exe 2022-10-13 10:02:20.000000
0xb88fcc30d390 UDPv4 0.0.0.0 49801 * 0 2128 svchost.exe 2022-10-13 10:08:50.000000
0xb88fcc30d390 UDPv6 :: 49801 * 0 2128 svchost.exe 2022-10-13 10:08:50.000000
0xb88fcc588b420 UDPv4 0.0.0.0 50103 * 0 2128 svchost.exe 2022-10-13 09:23:28.000000
0xb88fcc588b420 UDPv6 :: 50103 * 0 2128 svchost.exe 2022-10-13 09:23:28.000000
0xb88fcc34aa770 UDPv4 127.0.0.1 53335 * 0 2276 svchost.exe 2022-10-13 09:23:09.0000
00
0xb88fcc30c580 UDPv4 0.0.0.0 54648 * 0 2128 svchost.exe 2022-10-13 10:08:50.000000
0xb88fcc30c580 UDPv6 :: 54648 * 0 2128 svchost.exe 2022-10-13 10:08:50.000000
0xb88fcc589d50 UDPv6 fe80::e9d5:dbe2:72d9:5582 54859 * 0 5908 svchost.exe 2022-10-13 09:23:29.000000
13 09:23:29.000000
0xb88fcc58a2e40 UDPv6 ::1 54860 * 0 5908 svchost.exe 2022-10-13 09:23:29.000000
0xb88fcc58a3610 UDPv4 192.168.1.116 54861 * 0 5908 svchost.exe 2022-10-13 09:23:29.0000
00
0xb88fcc58a2800 UDPv4 127.0.0.1 54862 * 0 5908 svchost.exe 2022-10-13 09:23:29.0000
00
0xb88fcc30db60 UDPv4 0.0.0.0 61500 * 0 2128 svchost.exe 2022-10-13 10:08:50.000000
0xb88fcc30db60 UDPv6 :: 61500 * 0 2128 svchost.exe 2022-10-13 10:08:50.000000
0xb88fcc30d9d0 UDPv4 0.0.0.0 64087 * 0 2128 svchost.exe 2022-10-13 10:08:50.000000
0xb88fcc30d9d0 UDPv6 :: 64087 * 0 2128 svchost.exe 2022-10-13 10:08:50.000000

C:\Lab\Forensics>
```

Conclusion:

In a digital forensics investigation, extracting evidence from dumped volatile memory using the **Volatility** tool provides crucial insights into system processes, active network connections, running applications, and potential malicious activities. Information such as passwords, encryption keys, or running malware can be retrieved, aiding in reconstructing events or identifying suspects. By analyzing the volatile memory image, investigators can gather time- sensitive evidence that is critical to solving the case.